

Diseases: Causes and Types

What a disease is, what causes it, and the key split between germ-borne and lifestyle diseases.

Two big types of disease

A **disease** is a condition that affects the normal working of the body or mind. Disease-causing organisms (bacteria, viruses, fungi, protozoa, worms) are called **pathogens**.

	Communicable	Non-communicable
Cause	Pathogens (germs)	Lifestyle, diet, environment, deficiency
Spreads?	Yes, person to person	No
Examples	Typhoid, dengue, flu, chickenpox, COVID-19	Diabetes, heart disease, cancer, asthma

More terms

- **Deficiency diseases** (scurvy, anaemia, goitre) come from lack of a nutrient — they are non-communicable.
- **Chronic diseases** last a long time (more than 3 months), e.g. diabetes, asthma.
- **Parasites** (e.g. worms) live in/on another living being and feed on it.

How communicable diseases spread

- **Air** — coughing/sneezing (cold, flu, TB).
- **Contaminated water/food** (cholera, typhoid, hepatitis A).
- **Direct/indirect contact** (sharing personal items).
- **Vectors** — insects like mosquitoes/houseflies (malaria, dengue).

Note: non-communicable diseases (NCDs) now cause most deaths in India — due to processed food, less exercise and longer lives.

Where students lose marks

Trap 1 — Diabetes/asthma are NOT communicable. They are non-communicable (lifestyle), not caused by germs and do not spread.

Trap 2 — "Pathogen" = the germ, not the disease. Bacteria, viruses, fungi, protozoa and worms are pathogens.

Trap 3 — A vector is the carrier insect, not the germ. A mosquito is a vector; the malaria protozoan is the pathogen.

Deficiency diseases (scurvy, goitre) are non-communicable — caused by missing nutrients, not germs.

Model answers — write them like this

Example A. Classify: typhoid, diabetes, dengue, asthma.

Step 1: Caused by pathogens & spread → communicable.

Step 2: Lifestyle-linked, don't spread → non-communicable.

∴ Communicable: typhoid, dengue. Non-communicable: diabetes, asthma.

Example B. Why are mosquitoes called vectors?

Step 1: They carry pathogens from an infected person to a healthy one.

∴ A vector is an insect that spreads a pathogen, e.g. mosquito for malaria/dengue.

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